BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 075 · Validation and Authentication

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which integers can random(1,6) produce in OCR ERL?

Q2. What is HTTP used for?
A. Sending outgoing email
B. Syncing an email inbox
C. Transferring web resources
D. Looking up IP addresses

Q3. “Penetration testing permits anyone to attack any network without permission.” Correct this. Explain why.

Q4. A zero-indexed 2D array is [[11,12],[21,22]]. Give grid[1,0], then state the indexes needed to access 12.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Validation — Checks input against specified rules, rejecting values outside the allowed range.

Range check — Tests whether a numeric value is between minimum and maximum limits.

Authentication — Checks identity, for example using username and password, rather than checking input format.

Re-prompt — A valid design asks again when input does not meet a validation rule.

Test case — An input and expected outcome used to check correctness.

Bug — A defect in the program.

New learning

Validation: Checks input against specified rules, rejecting values outside the allowed range. Consider this in a online age entry.

Range check: Tests whether a numeric value is between minimum and maximum limits. Consider this in a exam score entry.

Authentication: Checks identity, for example using username and password, rather than checking input format. Consider this in a club registration.

Re-prompt: A valid design asks again when input does not meet a validation rule. Consider this in a account login.

Worked example

A online age entry needs validation and range check. Explain why. [4 marks]

1. Identify the inputs, constraints and expected result.

2. Validation [1]; Checks input against specified rules, rejecting values outside the allowed range. [1].

3. Range check [1]; Tests whether a numeric value is between minimum and maximum limits. [1].

4. Give a specific point and a linked reason for each feature.

We Do · Practise with guidance

Guided 1

In a exam score entry, explain the purpose of validation. [2 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Validation [1]; Checks input against specified rules, rejecting values outside the allowed range. [1]. Authentication [1]; Checks identity, for example using username and password, rather than checking input format. [1]. Re-prompt [1]; A valid design asks again when input does not meet a validation rule. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Any integer from 1 through 6 inclusive.

Q2. C — Transferring web resources

Q3. Testing with permission, within agreed limits, to find and fix security weaknesses.

Q4. (a) grid[1,0] is 21; 12 is at grid[0,1].

I Do / We Do pages · Guided practice

Q1. Validation [1]; Checks input against specified rules, rejecting values outside the allowed range. [1].

Q2. Range check [1]; Tests whether a numeric value is between minimum and maximum limits. [1].

Q3. Authentication [1]; Checks identity, for example using username and password, rather than checking input format. [1]. Re-prompt [1]; A valid design asks again when input does not meet a validation rule. [1].

You Do pages · Code First

```
octet = 256
if octet >= 0 AND octet <= 255 then
    print("Accepted")
else
    print("Rejected")
endif
```

Expected result: Rejected

You Do pages · Independent questions

Q1. Validation [1]; Authentication [1].

Q2. Range check [1]; Tests whether a numeric value is between minimum and maximum limits. [1]. Authentication [1]; Checks identity, for example using username and password, rather than checking input format. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Validation [1]; Checks input against specified rules, rejecting values outside the allowed range. [1].

Guided 2

A club registration needs range check. Explain its role. [2 marks]

Worked method: Range check [1]; Tests whether a numeric value is between minimum and maximum limits. [1].

Guided 3

A account login uses authentication and re-prompt. Explain both. [4 marks]

Worked method: Authentication [1]; Checks identity, for example using username and password, rather than checking input format. [1]. Re-prompt [1]; A valid design asks again when input does not meet a validation rule. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the validation expression so an octet of 256 is rejected.

```
octet = 256
if ... then
    print("Accepted")
else
    print("Rejected")
endif
```

write

Question 1

State TWO concepts relevant to this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 2

Explain how range check and authentication help in a school trip form. [4 marks] Source: Original OCR J277-style teaching question; not an official past paper.

Question 3

Explain the roles of validation, authentication and re-prompt in a café membership. [6 marks] Source: Original OCR J277-style teaching question; not an official past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Validation checks inputs against defined rules and can reject values outside the allowed range.